

The Emergence of Open Source Players in the Database Space

This introductory article takes a sweeping look at the current database domain. The rapid expansion of this domain over the past five years is fascinating and makes for interesting reading, especially since more and more open source DBs are becoming popular.



Technology has been changing rapidly over the years. If we look at the telecom space, we had barely been introduced to mobile phones in early 2000, and now we have so many variants of mobile phones. The third generation of mobile technology is behind us. In the computing space, the bulky PCs and laptops of 2005 have morphed into sleek quad-core phones and tablets. Software has also been keeping pace with these developments, such as the tremendous changes in the operating system space. However, software that has not changed for a long time is in the 'database' space. Until recently (say 2009), and probably for many people even today, the term 'database' implies RDBMS (relational database management system). The relational database is the de-facto standard of databases, and advancements in basic technology have evolved little over the past 40 years.

The Apache Foundation's Hadoop signalled the end of this era. Today, the database field is going overboard to catch up with other software domains, trying to make up for lost time. It would be reasonable to say that the database

domain is one of the interesting fields for a software professional to be in today.

What has triggered this change? The answer is best summarised by quoting Michael Stonebraker, the great DB scientist. "One size does not fit all," he said. The use case of a database has expanded tremendously in the past few years and trying to fit an RDBMS to all these use cases does not make sense. Let us look at some of the factors driving the evolution of the database.

1. Connectivity has played a major role in the changing DB usage. Ubiquitous Internet connectivity has meant access to a lot of websites and the emergence of social networks like Facebook. These need to host very large databases that demand high performance. At the same time, there is also the need for a less rigid and simple schema for databases.
2. With the increasing power of computing devices and the emergence of Big Data, a need to offer more 'real-timeness' has emerged. For example, there is immense demand for real-time analytics. These kinds